

Abstract

Formula One performance engineering is centred on understanding the mechanisms responsible for vehicle performance rather than simply measuring race results. Although the finishing order provides the final outcome of a race, it does not explain how performance was generated or which engineering factors contributed to that result. Telemetry enables engineers to investigate these mechanisms by analysing driver inputs, observable vehicle-speed behaviour and lap-time evolution throughout a race weekend.

This report presents a telemetry-based engineering investigation of the 2026 Austrian Grand Prix, focusing on the race performances of George Russell and Andrea Kimi Antonelli. Publicly available telemetry obtained through FastF1 was analysed using Python to examine race pace evolution, sector performance, corner behaviour and driver inputs. The investigation follows a structured engineering methodology designed to separate objective observations from engineering interpretations while explicitly recognising the limitations imposed by unavailable telemetry channels.

Rather than attempting to replicate proprietary Formula One analyses, the objective of this project is to demonstrate how publicly available data can be transformed into evidence-supported engineering conclusions through a systematic analytical framework. Key engineering findings are supported by the available telemetry and interpreted within an explicit confidence framework that distinguishes directly observed evidence from engineering hypotheses.

The investigation demonstrates that meaningful engineering insights can be extracted from public telemetry despite the absence of several important vehicle channels. However, conclusions regarding vehicle balance, differential behaviour, tyre condition and steering characteristics remain limited by the available data and are therefore presented only as engineering hypotheses where appropriate.

This report establishes a repeatable methodology for telemetry-based race investigations that can be applied consistently across future Formula One events. Beyond analysing a single race, the project aims to develop a structured engineering workflow that prioritises evidence-based reasoning, transparency and technical accuracy throughout the investigation process.

Keywords: Formula One, Performance Engineering, Telemetry Analysis, Race Engineering, FastF1, Motorsport Engineering, Vehicle Dynamics.

1. Introduction

1.1 Background

Formula One represents one of the most data-intensive engineering environments in modern motorsport. During every race weekend, thousands of telemetry channels are recorded continuously, allowing engineers to evaluate driver performance, vehicle behaviour, tyre usage, aerodynamic efficiency and overall race execution. These data support engineering decisions before, during and after each session and play a fundamental role in maximising competitive performance.

Within a Formula One team, telemetry contributes to performance investigation by allowing engineers to compare driver inputs, vehicle behaviour and timing outcomes across different operating conditions.

Publicly available telemetry provides only a limited subset of these channels. Nevertheless, when analysed using an appropriate engineering methodology, it remains possible to investigate many aspects of race performance while acknowledging the limitations created by unavailable data.

1.2 Motivation

Most publicly available Formula One analyses focus primarily on visual comparisons of lap times, telemetry traces or race statistics. While these analyses are valuable for describing race events, they rarely follow the structured reasoning process adopted by professional performance engineers.

The motivation for this project is to develop a repeatable engineering methodology capable of transforming publicly available telemetry into evidence-supported engineering conclusions. Rather than simply describing performance differences, this investigation seeks to evaluate the engineering interpretations supported by those differences while clearly distinguishing between confirmed observations and hypotheses requiring further validation.

This approach also provides an opportunity to develop analytical skills directly relevant to motorsport performance engineering by combining telemetry interpretation, vehicle dynamics principles and structured engineering reasoning within a single investigation framework.

1.3 Existing Public Telemetry Approaches

In recent years, publicly available Formula One telemetry has become increasingly accessible through platforms such as FastF1 and the official FIA timing feed. This has enabled engineers, researchers and motorsport enthusiasts to perform increasingly sophisticated analyses of race performance using open-source data.

Most publicly available investigations focus on visual comparisons of telemetry channels, including speed traces, throttle application, braking zones and racing lines. These analyses successfully identify where performance differences occur but do not always distinguish clearly between directly observed telemetry differences and interpretations of their possible underlying mechanisms.

Previous work has also tended to prioritise descriptive observations rather than structured engineering reasoning. As a result, engineering conclusions are often presented without explicitly

distinguishing between directly measured evidence and engineering interpretation. This limits transparency and makes it difficult to evaluate the reliability of the findings.

The present investigation addresses these limitations by combining telemetry analysis with a structured engineering methodology in which observations, evidence, engineering interpretations and confidence assessments are explicitly separated throughout the report.

1.4 Project Rationale

This project addresses a practical limitation of public telemetry analysis: measured performance differences can often be identified more confidently than their underlying physical causes. The investigation therefore develops a transparent workflow in which observations, engineering interpretations, data limitations and confidence levels are treated separately.

The methodology is designed to remain repeatable across future race investigations while preventing unavailable telemetry from being replaced by unsupported assumptions.

1.5 Contributions of this Investigation

The primary contributions of this investigation are summarised below:

- Development of a structured telemetry-based engineering workflow.
- Introduction of an explicit engineering confidence framework.
- Separation of objective observations from engineering interpretations.
- Definition of a consistent set of performance metrics for race- and corner-level comparison.
- Establishment of a methodology applicable to future Formula One race investigations using publicly available telemetry.

1.6 Research Objectives

Primary Objective

To investigate measurable race-performance differences between George Russell and Andrea Kimi Antonelli during the 2026 Austrian Grand Prix and evaluate the engineering interpretations supported by publicly available telemetry.

Secondary Objectives

- Investigate race pace evolution throughout the event.
- Identify the sectors and corners contributing most significantly to race performance.
- Evaluate observable braking-state, speed and throttle characteristics through selected corner sequences.
- Compare driver behaviour using telemetry-derived evidence.
- Distinguish confirmed engineering observations from engineering hypotheses.
- Establish a repeatable methodology for future race investigations.

1.7 Scope

This investigation is based exclusively on publicly available telemetry and timing information obtained through FastF1 and FIA timing data.

The analysis includes:

- Lap and sector timing
- Vehicle speed
- Throttle position
- Binary brake application
- Gear selection
- GPS-derived distance and position
- Race position and tyre-compound information

The following channels are unavailable and therefore limit the investigation:

- Steering angle
- Continuous brake pressure
- Differential state
- Tyre temperature and pressure
- Detailed tyre condition
- Suspension state
- Yaw rate

Consequently, conclusions regarding vehicle balance and certain aspects of tyre and differential behaviour are treated as hypotheses rather than confirmed engineering findings.

1.8 Engineering Philosophy

The purpose of this investigation is not to determine which driver performed better but to understand why measurable performance differences occurred.

To achieve this objective, every section of this report follows a structured engineering workflow:

1. Define the engineering question.
2. Validate the available telemetry.
3. Record objective observations.
4. Support observations using multiple sources of evidence.
5. Develop engineering interpretations.
6. Evaluate alternative hypotheses.
7. Assign an appropriate confidence level.
8. Recommend future engineering investigations where necessary.

By separating observations from interpretations and explicitly acknowledging data limitations, the investigation aims to maintain technical transparency while avoiding conclusions that cannot be justified by the available evidence.

Chapter 2. Methodology

2.1 Research Methodology

This investigation follows a structured engineering methodology designed to transform publicly available telemetry into evidence-supported engineering conclusions. Rather than analysing individual telemetry channels independently, the methodology combines multiple data sources to investigate race performance from an engineering perspective.

The investigation begins by defining a clear engineering question before analysing telemetry data. Each observation is then supported using multiple telemetry channels where possible. Engineering interpretations are developed only after objective observations have been established, while alternative explanations are considered throughout the analysis to reduce subjective interpretation.

Finally, key engineering conclusions are evaluated according to the confidence supported by the available evidence.

This methodology is intended to improve transparency, repeatability and technical consistency throughout the investigation.

2.2 Data Sources

The investigation relies exclusively on publicly available Formula One timing and telemetry information.

The primary data source is the FastF1 Python library, which provides access to official Formula One timing information published during race weekends. FastF1 combines FIA timing feeds with circuit metadata and telemetry channels, allowing engineers and researchers to perform detailed race analyses using open data.

Telemetry channels used throughout this report include:

- Vehicle Speed
- GPS Position
- Throttle Position
- Brake Application
- Gear Selection
- Sector Timing
- Lap Timing

All data processing and analysis were performed using Python.

Source	Purpose
FastF1	Telemetry extraction and timing
FIA timing feed	Official timing data
Python	Data processing
Pandas	Data analysis
NumPy	Numerical calculations

Matplotlib	Visualisation
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Table 2.1 Data Sources

2.3 Data Processing

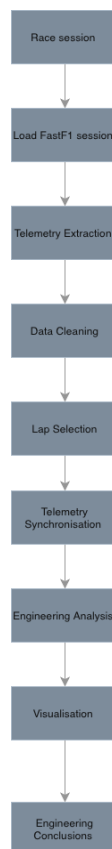
Raw telemetry cannot be analysed directly. Several preprocessing steps were therefore performed before engineering analysis.

The processing workflow consisted of:

1. Loading race session data.
2. Selecting representative race laps.
3. Synchronising telemetry channels.
4. Removing invalid or incomplete laps where necessary.
5. Calculating sector and corner performance.
6. Producing engineering visualisations

Representative race laps were selected to reduce the influence of pit-entry, pit-exit and major non-representative laps. This filtering improves comparability but does not create fully controlled like-for-like operating conditions between the drivers.

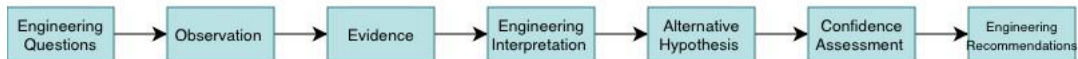
These preprocessing steps ensure that comparisons between drivers are performed using consistent datasets.



2.4 Engineering Workflow

Unlike descriptive telemetry analysis, this investigation follows a structured engineering workflow.

Every chapter uses the following sequence:



This structure ensures that engineering conclusions remain directly linked to supporting telemetry rather than subjective judgement.

2.5 Engineering Questions

The investigation is guided by the following research questions:

ID	Research Question	Output
Q1	How did representative race pace evolve?	Pace Investigation
Q2	Which sectors generated performance?	Sector Analysis
Q3	Which corners generated performance?	Corner Investigation
Q4	Which interpretations are supported by telemetry?	Engineering Discussion
Q5	Which conclusions are supported by evidence?	Confidence Assessment
Q6	Which hypotheses remain uncertain?	Limitations
Q7	Which additional data are required?	Future Work

Table 2.2 Engineering Questions

2.6 Data Limitations

Although publicly available telemetry provides valuable engineering information, several important vehicle channels are unavailable.

These include:

- Steering Angle
- Continuous Brake Pressure
- Differential State
- Wheel Speed

- Tyre Temperature
- Tyre Pressure
- Suspension Travel
- Yaw Rate

Consequently, direct assessment of vehicle balance, tyre behaviour and differential operation is beyond the scope of this investigation.

Whenever engineering reasoning requires unavailable telemetry, conclusions are explicitly presented as hypotheses rather than confirmed findings.

Channel	Available	Engineering Use / Limitation
GPS Position / Distance	✓	Spatial alignment and corner localisation
Speed	✓	Primary corner-performance evidence
Binary Brake State	✓	Brake-application timing
Throttle Position	✓	Throttle-transition analysis
Gear	✓	Contextual driver-input evidence
Steering Angle	✗	Steering behaviour cannot be determined
Tyre Temperature / Pressure	✗	Detailed tyre state cannot be determined
Differential State	✗	Differential behaviour unavailable
Continuous Brake Pressure	✗	Brake-force modulation unavailable

Table 2.3 Telemetry Availability

2.7 Confidence Framework

Engineering confidence is determined by the **directness, consistency and limitations of the available evidence**, rather than by telemetry-channel count alone.

High Confidence

The finding is directly observable in the available timing or telemetry data and does not require an unmeasured physical mechanism to be assumed.

Medium Confidence

The available telemetry supports a reasonable engineering interpretation, but additional vehicle-state or driver-input information would be required for confirmation.

Low Confidence

The interpretation relies substantially on indirect evidence or on variables that are not directly measured.

Not Directly Observable

The available public dataset does not provide sufficient information to determine the mechanism reliably.

Level	Definition	Engineering Meaning
High	Directly supported by available evidence	Observation supported
Medium	Evidence supports an interpretation but additional data would improve certainty	Reasonably supported interpretation
Low	Relies substantially on indirect evidence	Engineering hypothesis
Not Directly Observable	Required information is unavailable	No reliable mechanism-level conclusion

Table 2.4 Confidence Classification

2.8 Statistical Analysis

Race performance cannot be evaluated solely by comparing individual lap times. Instead, statistical methods are employed to investigate overall pace consistency, performance evolution and variation throughout the race.

Several complementary analytical techniques are used throughout this investigation. Scatter plots visualise lap time distribution, moving averages reduce short-term fluctuations to reveal long-term pace trends, while linear trend analysis assists in identifying the progression of race pace during individual stints.

These methods reduce the influence of isolated observations and provide complementary descriptions of representative pace, variability and race-phase evolution.

2.9 Derived Metrics and Numerical Limitations

Public telemetry allows additional quantities to be estimated from measured channels. However, derived variables introduce additional uncertainty because they depend on telemetry sampling, synchronisation, interpolation and numerical differentiation.

Derived quantities are therefore used only where their numerical behaviour is sufficiently stable for the intended analysis. They are not treated as equivalent to directly measured team-level vehicle channels.

2.9.1 Longitudinal Acceleration

Vehicle longitudinal acceleration can be estimated from:

$$a_x = dv/dt$$

where v is vehicle speed and t is time.

However, numerical differentiation of sampled public speed data can produce large instantaneous spikes, particularly around rapid speed changes or irregular sampling intervals. Within this investigation, the derived acceleration trace is therefore not used as a primary quantitative measure of braking force or physical vehicle acceleration.

Where referenced, it is treated only as a qualitative indication of the timing and shape of speed-change events. Quantitative braking and corner diagnosis instead prioritises directly available speed, binary brake state, throttle position and cumulative time delta.

2.10 Engineering Evidence Hierarchy

Evidence Type	Interpretation
Direct timing / telemetry	Primary quantitative evidence
Derived metrics	Supporting evidence subject to numerical validation
Statistical analysis	Evidence of distribution, trend and repeatability
Engineering interpretation	Evidence-supported explanation requiring judgement
Unmeasured mechanism	Hypothesis requiring additional data

To maintain transparency, engineering conclusions are classified according to the strength of the supporting evidence.

This hierarchy is applied throughout the report to ensure that conclusions remain proportional to the available evidence.

2.11 Chapter Summary

This chapter established the engineering methodology used throughout the investigation. By combining telemetry preprocessing, structured engineering reasoning and explicit confidence assessment, the analysis aims to produce conclusions that are technically transparent, repeatable and supported by available evidence. The following chapters apply this methodology to the Austrian Grand Prix race data, beginning with the overall race context before progressing to detailed pace, sector and corner investigations.

CHAPTER 3.Austrian Grand Prix Event Overview

3.1 Circuit Characteristics

The Red Bull Ring is one of the shortest circuits on the Formula One calendar, measuring approximately 4.3 kilometres in length. Despite its relatively simple layout, the circuit presents several distinct engineering challenges due to its significant elevation changes, heavy braking zones and traction-limited corner exits.

The combination of long acceleration zones and varied corner types makes both corner execution and subsequent speed development relevant to lap performance.

From an engineering perspective, the Red Bull Ring requires an effective compromise between straight-line efficiency and mechanical grip. Performance therefore depends on managing the compromise between straight-line efficiency and effective corner execution across the circuit's different speed ranges.

Characteristic	Engineering Relevance
Short Lap Length	Small time differences can represent meaningful performance differences
Elevation Changes	Create varied braking and acceleration phases around the lap
Long Acceleration Zones	Increase the importance of post-corner speed development
Major Braking Zones	Make braking-phase execution relevant to corner performance
Varied Corner Speeds	Require performance to be evaluated across different corner types
Linked Corner Sequences	Require local gains to be assessed together with downstream effects

Figure 3.1 Circuit Engineering Characteristics

3.2 Engineering Challenges

The Red Bull Ring combines significant braking events with several acceleration-sensitive corner sequences, making both braking-phase execution and downstream speed development relevant to the investigation.

The primary engineering priorities for this event include:

- Maximising post-corner speed development.
- Achieving effective execution through major braking phases.
- Preserving tyre performance over race distance.

- Achieving repeatable race pace across multiple stints.
- Minimising lap time variation throughout the race.

These priorities define the engineering focus of the present investigation.

Priority	Importance	Investigation Focus
Post-Corner Speed	High	Exit / Downstream Speed
Braking Phase	High	Binary Brake State + Speed Trace
Race Repeatability	High	Representative Lap-Time Variability
Corner Performance	High	Corner Delta
Driver Inputs	Medium	Throttle Transition + Binary Brake State
Engineering Interpretation	Medium	Evidence-Supported Hypotheses

Table 3.1 Primary Engineering Priorities

3.3 Circuit Corner Classification

To structure the corner-level investigation, selected corners are classified according to their primary analysis focus and the telemetry metrics used for comparison.

Turn	Analysis Focus	Primary KPI	Secondary KPI
T1	Corner sequence	Corner delta	Minimum speed
T3	Heavy braking / low speed	Corner delta	Minimum speed
T4	Acceleration-sensitive	Corner delta	Throttle transition
T6	Medium-speed sequence	Corner delta	Minimum speed / throttle transition
T7	Linked sequence	Corner delta	T6–T7 combined delta
T9	High-speed sequence	Corner delta	Minimum speed
T10	Final corner	Corner delta	Throttle transition / downstream speed

This classification provides the framework used throughout the corner-by-corner investigation presented later in the report.

3.4 Candidate Performance Factors

Before analysing telemetry, several observable performance factors are identified as investigation targets for the Red Bull Ring.

These factors include:

- braking-phase behaviour;
- speed retention;
- minimum speed;
- throttle transition;
- exit and downstream speed;
- corner-sequence interaction;
- race-level repeatability.

These factors are treated as investigation targets rather than assumed causes of performance differences. Where the required telemetry is unavailable, the analysis does not attempt to establish the underlying physical mechanism.

3.5 Investigation Assumptions

1. Public telemetry provides a sufficiently consistent basis for the relative comparisons performed within the scope of this investigation.
2. Missing telemetry channels limit engineering certainty.
3. Observations are separated from engineering interpretations.
4. Engineering conclusions require supporting evidence.
5. Multiple hypotheses are considered where the available evidence does not establish a unique mechanism.

These assumptions are intended to improve transparency and minimise subjective interpretation.

Assumption	Purpose
Evidence precedes interpretation	Reduce subjective judgement
Multiple telemetry channels support conclusions	Increase reliability
Missing data reduce confidence	Prevent unsupported claims
Alternative hypotheses are evaluated	Improve engineering reasoning
Recommendations are evidence-based	Maintain technical validity

Table 3.2 Engineering Assumptions

3.6 Engineering Objectives for the Austrian GP Investigation

The investigation aims to answer the following engineering questions:

- How did representative race pace evolve?
- Which sectors contributed most significantly to overall performance?
- Which corners produced the largest lap time differences?
- During which phases of selected corners did measurable time differences develop?
- Which conclusions are directly supported by telemetry?
- Which findings require additional vehicle data before confirmation?

These questions guide every analytical section that follows.

3.7 Engineering Key Performance Indicators

To enable systematic comparison between drivers, a set of engineering Key Performance Indicators (KPIs) is defined for each phase of the investigation. These KPIs represent measurable performance characteristics extracted directly or indirectly from the available telemetry.

KPI	Engineering Purpose
Lap Time	Representative overall pace
Sector Time	Sector-level performance
Minimum Speed	Corner-speed characteristic
Exit / Downstream Speed	Post-corner speed development
Binary Brake State	Brake-application timing
Throttle Transition	Driver-input progression
Corner Delta	Relative selected-corner performance
Pace Variability	Representative-lap repeatability

These KPIs form the basis of all subsequent race and corner analyses.

3.8 Investigation Roadmap

The investigation progresses from overall race performance towards increasingly detailed engineering analysis.

The sequence begins with race pace evaluation before examining sector performance, individual corner behaviour and finally the engineering interpretations supported by the observed telemetry.

This hierarchical structure ensures that engineering conclusions remain supported by progressively stronger evidence while maintaining consistency throughout the investigation.

3.9 Chapter Summary

The Austrian Grand Prix presents a combination of significant braking events, varied corner types and long acceleration phases that make it suitable for telemetry-based performance investigation. This chapter established the circuit context, candidate performance factors and measurable KPIs used throughout the subsequent analysis.

The investigation now progresses from race-level observations towards increasingly detailed performance diagnosis. Chapter 4 begins with race progression and representative pace before the analysis moves towards sector- and corner-level performance differences in the subsequent chapters.